Factorising with Common Factors

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KEY WORDS factorise *(whakatauwehe)* · factor *(tauwehe)* · highest common factor *(tauwehe pātahi nui rawa)*

A A number as the common factor ●○○

Remember: Factorising is the reverse of expanding. Take out the largest number that divides every term: $6x + 9 = 3(2x + 3)$.

1. Write down the highest common factor of each pair of terms.

a) $6x$ and 9 _____ **b)** $10a$ and 15 _____ **c)** $8y$ and 12 _____ **d)** $14m$ and 21 _____

2. Factorise fully.

a) $2x + 6$ = _____ **b)** $3x + 12$ = _____ **c)** $5x - 10$ = _____ **d)** $6x + 9$ = _____

e) $8x - 4$ = _____ **f)** $6x + 12$ = _____ **g)** $14x + 21$ = _____ **h)** $15x - 20$ = _____

i) $8x - 24$ = _____ **j)** $18x + 9$ = _____ **k)** $30x - 20$ = _____ **l)** $12x + 48$ = _____

m) $15x + 30$ = _____ **n)** $30x - 42$ = _____ **o)** $40x + 60$ = _____ **p)** $25x - 100$ = _____

B A letter as the common factor ●●○

Remember: If every term has the same letter, take the letter out: $x^2 + 5x = x(x + 5)$.

3. Factorise.

a) $x^2 + 5x$ = _____ **b)** $x^2 - 3x$ = _____ **c)** $2x^2 + 7x$ = _____ **d)** $x^2 + 9x$ = _____

e) $3x^2 - x$ = _____ **f)** $x^2 - 12x$ = _____ **g)** $5x^2 + 2x$ = _____ **h)** $4x^2 - 9x$ = _____

i) $x^2 + x$ = _____ **j)** $7x^2 - 3x$ = _____ **k)** $x^2 - 20x$ = _____ **l)** $9x^2 + 4x$ = _____

C A number and a letter ●●○

Watch out: Take out everything that is common. $4x^2 + 6x$ is $2x(2x + 3)$, not just $2(2x^2 + 3x)$.

4. Factorise fully.

a) $2x^2 + 6x$ **b)** $3x^2 - 6x$ **c)** $4x^2 + 8x$ **d)** $4x^2 + 6x$

e) $5x^2 - 15x$ **f)** $6x^2 + 15x$ **g)** $6x^2 + 6x$ **h)** $12x^2 - 8x$

i) $7x^2 + 14x$ **j)** $10x^2 - 6x$ **k)** $9x^2 - 9x$ **l)** $10x^2 + 15x$

m) $10x^2 + 30x$ **n)** $12x^2 - 15x$ **o)** $16x^2 + 8x$ **p)** $12x^2 - 18x$

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D Two letters ●●●

5. Factorise fully.

a) $6ab + 9a$ **b)** $2xy + 8x$ **c)** $5mn - 10m$

d) $4ab + 8ac$ **e)** $xxy + xyy$ **f)** $9pq - 3p$

g) $12aa + 18ab$ **h)** $6xxy - 8xyy$ **i)** $14hk - 21k$

j) $10xx + 15xy$ **k)** $8pq - 16pr$ **l)** $10aab + 20abb$

E Mixed practice ●●●

⚠ **Watch out:** Check your answer by expanding it. You should get back to the expression you started with.

6. Factorise fully.

a) $4x + 20$ **b)** $x^2 + 8x$ **c)** $9x^2 - 6x$ **d)** $6ab + 2a$

e) $18x - 45$ **f)** $5x^2 + 20x$ **g)** $7x - 14y$ **h)** $6x^2 - x$

i) $24x + 36$ **j)** $8mn + 20m$ **k)** $24x^2 - 8x$ **l)** $xxy - xyy$

★ Extension

7. These factorise into two brackets. Find two numbers that multiply to give the last number and add to give the number in front of x.

a) $x^2 + 5x + 6$ **b)** $x^2 + 6x + 5$ **c)** $x^2 + 7x + 12$

d) $x^2 + 8x + 12$ **e)** $x^2 - 5x + 6$ **f)** $x^2 + 3x - 10$

g) $x^2 + 8x + 7$ **h)** $x^2 - 8x + 15$ **i)** $x^2 + 7x - 8$

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Factorising with Common Factors

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS factorise *(whakatauwehe)* · factor *(tauwehe)* · highest common factor *(tauwehe pātahi nui rawa)*

A A number as the common factor ●○○

Remember: Factorising is the reverse of expanding. Take out the largest number that divides every term: $6x + 9 = 3(2x + 3)$.

1. Write down the highest common factor of each pair of terms.

a) $4x$ and 6 _____ **b)** $12a$ and 18 _____ **c)** $9y$ and 15 _____ **d)** $20m$ and 30 _____

2. Factorise fully.

a) $2x + 10$ = _____ **b)** $3x + 6$ = _____ **c)** $4x - 12$ = _____ **d)** $6x + 10$ = _____

e) $10x - 5$ = _____ **f)** $6x + 18$ = _____ **g)** $21x + 14$ = _____ **h)** $12x - 20$ = _____

i) $9x - 18$ = _____ **j)** $16x + 8$ = _____ **k)** $20x - 30$ = _____ **l)** $11x + 44$ = _____

m) $12x + 60$ = _____ **n)** $30x - 6$ = _____ **o)** $60x + 40$ = _____ **p)** $30x - 120$ = _____

B A letter as the common factor ●●○

Remember: If every term has the same letter, take the letter out: $x^2 + 5x = x(x + 5)$.

3. Factorise.

a) $x^2 + 7x$ = _____ **b)** $x^2 - 4x$ = _____ **c)** $2x^2 + 5x$ = _____ **d)** $x^2 + 8x$ = _____

e) $3x^2 - 2x$ = _____ **f)** $x^2 - 15x$ = _____ **g)** $5x^2 + 3x$ = _____ **h)** $4x^2 - 7x$ = _____

i) $x^2 - x$ = _____ **j)** $7x^2 - 2x$ = _____ **k)** $x^2 - 25x$ = _____ **l)** $8x^2 + 5x$ = _____

C A number and a letter ●●○

Watch out: Take out everything that is common. $4x^2 + 6x$ is $2x(2x + 3)$, not just $2(2x^2 + 3x)$.

4. Factorise fully.

a) $2x^2 + 10x$ **b)** $3x^2 - 12x$ **c)** $5x^2 + 10x$ **d)** $6x^2 + 4x$

e) $4x^2 - 12x$ **f)** $9x^2 + 15x$ **g)** $8x^2 + 8x$ **h)** $8x^2 - 12x$

i) $6x^2 + 30x$ **j)** $6x^2 - 10x$ **k)** $7x^2 - 7x$ **l)** $15x^2 + 10x$

m) $10x^2 + 70x$ **n)** $15x^2 - 12x$ **o)** $18x^2 + 9x$ **p)** $18x^2 - 12x$

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D Two letters ●●●

5. Factorise fully.

a) $6ab + 10a$ **b)** $3xy + 6x$ **c)** $4mn - 12m$

d) $5ab + 15ac$ **e)** $aab + abb$ **f)** $10pq - 2p$

g) $12aa + 8ab$ **h)** $6xxy - 15xyy$ **i)** $18hk - 45k$

j) $12xx + 8xy$ **k)** $6pq - 18pr$ **l)** $20aab + 10abb$

E Mixed practice ●●●

⚠ **Watch out:** Check your answer by expanding it. You should get back to the expression you started with.

6. Factorise fully.

a) $5x + 20$ **b)** $x^2 + 6x$ **c)** $12x^2 - 8x$ **d)** $6ab + 3a$

e) $16x - 40$ **f)** $7x^2 + 14x$ **g)** $5x - 15y$ **h)** $5x^2 - x$

i) $28x + 42$ **j)** $12mn + 18m$ **k)** $18x^2 - 9x$ **l)** $aab - abb$

★ Extension

7. These factorise into two brackets. Find two numbers that multiply to give the last number and add to give the number in front of x.

a) $x^2 + 5x + 4$ **b)** $x^2 + 7x + 10$ **c)** $x^2 + 8x + 15$

d) $x^2 + 9x + 14$ **e)** $x^2 - 5x + 4$ **f)** $x^2 + 4x - 12$

g) $x^2 + 10x + 9$ **h)** $x^2 - 8x + 12$ **i)** $x^2 + 6x - 7$

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